

Wardle Editorial Operating System (WEOS)

Part I — Vision & Editorial Philosophy

1. Purpose

The Wardle Editorial Operating System (WEOS) is the clinical editorial foundation that governs the creation, review, publication, and continuous improvement of diagnostic knowledge within Wardle.

WEOS exists to ensure that every educational artifact exposed to learners—from a single clue in a clinical case to a complex network of diagnostic relationships—is clinically defensible, educationally intentional, internally consistent, and governed through transparent editorial processes.

Wardle is not a content management system.

Wardle is not an AI writing platform.

Wardle is not a collection of isolated clinical cases.

Wardle is an operating system for governing diagnostic knowledge.

Every diagnosis, case, teaching objective, clinical pearl, differential diagnosis, reasoning pathway, and knowledge relationship forms part of a living editorial ecosystem that evolves through continuous review rather than one-time publication.

The purpose of WEOS is therefore not merely to store medical content, but to provide the structures, workflows, and governance required to build and maintain a trustworthy body of clinical knowledge at institutional scale.

2. Editorial Philosophy

Medical education is fundamentally different from conventional publishing.

Clinical knowledge cannot be considered complete simply because it has been written, nor can educational quality be measured by the quantity of available content.

Every published statement influences how future clinicians recognise disease, interpret evidence, construct differential diagnoses, and make decisions that ultimately affect patient care.

For this reason, WEOS treats every editorial artifact as a clinical decision rather than a document.

A teaching objective represents an educational commitment.

A diagnostic clue represents an instructional decision.

A knowledge relationship represents a clinical assertion.

A clinical pearl represents an expert recommendation.

Publication therefore signifies institutional confidence in the educational validity of these decisions rather than completion of a writing task.

Editorial work is consequently viewed as an ongoing process of refinement.

Knowledge is expected to evolve.

Evidence changes.

Clinical practice develops.

Educational priorities shift.

The editorial system must therefore remain capable of continuously improving existing knowledge while preserving accountability for every published decision.

3. Core Editorial Principles

The design of WEOS is guided by several enduring principles.

3.1 Knowledge Before Content

WEOS governs knowledge rather than documents.

Individual pieces of content exist only to communicate clinical knowledge.

Knowledge therefore remains the primary asset of the system, while cases, explanations, objectives, and educational materials are different expressions of that knowledge.

3.2 Editorial Decisions Must Be Explicit

Every meaningful change to educational knowledge should correspond to an identifiable editorial decision.

The system should never silently transform educational content.

Editors should always understand:

- what changed
- why it changed
- who approved it
- what evidence supports it
- what educational impact the change produces

Editorial governance depends upon transparent decision-making rather than hidden automation.

3.3 Clinical Accuracy Is Necessary but Not Sufficient

Correct medical facts alone do not produce effective education.

Every artifact should also be evaluated according to its educational quality.

WEOS therefore governs:

- clinical correctness
- diagnostic reasoning
- educational usefulness
- learner progression
- curriculum alignment
- consistency with surrounding knowledge

Publication requires all of these dimensions to be considered together.

3.4 Every Artifact Has a Purpose

No editorial artifact exists in isolation.

Every artifact should answer a clearly defined educational question.

Examples include:

- Why should this teaching objective exist?
- Which misconception does this clinical pearl address?
- Which diagnostic uncertainty does this graph relationship resolve?
- Which reasoning step does this clue teach?
- Which learning outcome does this case support?

Artifacts without educational purpose create complexity without improving learning.

3.5 Human Judgment Governs Clinical Knowledge

Artificial intelligence can assist editorial work but cannot replace clinical responsibility.

The responsibility for educational quality, clinical accuracy, and publication always remains with qualified human reviewers.

AI may recommend.

AI may draft.

AI may analyse.

AI may identify inconsistencies.

Only human editorial governance establishes institutional trust.

3.6 Continuous Improvement Over Static Publication

Publication is not the end of editorial work.

Published knowledge should continuously evolve as:

- evidence changes
- guidelines change
- learner performance identifies weaknesses
- reviewers improve explanations
- new diagnostic relationships emerge
- curriculum priorities evolve

Every published artifact should therefore remain reviewable throughout its lifetime.

3.7 Editorial Consistency Is a System Property

Consistency should not depend upon individual reviewers remembering previous decisions.

The editorial system itself should promote consistency by exposing:

- conflicting knowledge
- duplicated teaching
- inconsistent terminology
- contradictory reasoning

- overlapping educational objectives
- missing educational coverage

Editorial quality should emerge from the system as much as from individual expertise.

4. Editorial Operating Model

WEOS views editorial work as a governed progression of clinical knowledge.

Rather than treating publication as a single event, knowledge moves through a sequence of increasingly trustworthy states.

Knowledge begins as an idea.

It develops into structured educational content.

It undergoes expert review.

It becomes institutional knowledge.

It continues to evolve throughout its published life.

The operating model therefore emphasises progression rather than completion.

Every editorial activity contributes to increasing confidence in the quality, educational value, and clinical reliability of knowledge.

This progression applies equally to diagnoses, cases, teaching objectives, clinical pearls, reasoning pathways, educational resources, and future editorial artifacts.

5. Scope of the Editorial System

WEOS governs every educational artifact that contributes to diagnostic learning.

These include, but are not limited to:

- Diagnoses
- Clinical cases
- Diagnostic clues
- Teaching objectives
- Teaching rules
- Clinical pearls

- Clinical pitfalls
- Differential diagnoses
- Diagnostic reasoning pathways
- Knowledge relationships
- Educational summaries
- Investigation guidance
- Management guidance
- Recall prompts
- Scoring systems
- References
- AI-generated editorial drafts
- Editorial reviews
- Publication decisions

The operating system intentionally separates the governance of these artifacts from the technology used to store or present them.

The editorial model should remain valid regardless of future implementation details.

6. Definition of Editorial Success

WEOS does not measure success by the number of published cases.

Instead, editorial success is defined by the quality of the governed knowledge.

An effective editorial system should consistently produce knowledge that is:

- Clinically accurate
- Educationally purposeful
- Internally consistent
- Evidence-informed
- Transparently governed
- Continuously improvable
- Explainable to reviewers
- Trustworthy to learners
- Scalable across specialties
- Sustainable for long-term maintenance

These characteristics form the standard against which every future capability within WEOS should be evaluated.

Whenever architectural decisions are made, the preferred solution should be the one that most effectively strengthens these characteristics.

Closing Principle

Wardle does not manage medical content.

Wardle governs clinical knowledge.

Every architectural decision within the Wardle Editorial Operating System should reinforce that principle.

Part II — The Editorial Domain Model

1. Purpose

The Editorial Domain Model defines the knowledge entities governed by the Wardle Editorial Operating System (WEOS).

These entities are not implementation objects, database tables, or software components. They are editorial concepts that represent how clinical knowledge is created, evaluated, maintained, and published.

Every capability within WEOS operates upon one or more of these entities.

The domain model establishes:

- what exists within the editorial system
- why it exists
- how entities relate to one another
- how knowledge progresses through editorial governance
- where editorial decisions occur

All future implementations should preserve these concepts regardless of the underlying technology.

2. Editorial Philosophy of the Domain

Medical knowledge is not published as isolated documents.

Instead, it exists as an interconnected network of educational artifacts that collectively teach diagnostic reasoning.

A diagnosis is not merely accompanied by a case.

It is explained through objectives.

It is reinforced through teaching rules.

It is clarified through clinical pearls.

It is distinguished through differential diagnoses.

It is connected through knowledge relationships.

It is evaluated through evidence.

It is continuously improved through editorial review.

WEOS therefore models editorial knowledge as an ecosystem rather than a hierarchy of documents.

3. Editorial Artifact Hierarchy

The editorial system governs a collection of related artifacts.

These artifacts exist at different levels of abstraction.

Clinical Knowledge

|

|— Diagnosis

| |— Metadata

| |— Educational Framework

| |— Clinical Cases

| |— Knowledge Relationships

| |— Educational Content

| |— Evidence

| |— Reviews

| |— Publication

|

|— Editorial Governance

The Diagnosis is the primary unit of editorial governance.

All other artifacts either describe, teach, validate, or govern the diagnosis.

4. Primary Editorial Entities

4.1 Diagnosis

The Diagnosis is the central knowledge entity within WEOS.

Every editorial activity ultimately contributes to improving one or more diagnoses.

A diagnosis defines the educational identity around which all associated knowledge is organised.

A diagnosis is responsible for:

- clinical identity
- taxonomy
- metadata
- educational scope
- publication status
- knowledge relationships
- associated educational artifacts

A diagnosis is never considered "finished."

It continuously evolves throughout its editorial lifetime.

4.2 Clinical Case

Clinical cases are educational expressions of a diagnosis.

Their purpose is not to document patient encounters but to teach diagnostic reasoning.

A case should progressively guide learners toward a diagnosis while reinforcing important discriminators, avoiding premature diagnostic disclosure, and supporting educational objectives.

Cases exist to teach.

They are not the diagnosis itself.

Multiple cases may exist for the same diagnosis.

Different cases may target different learner levels, clinical settings, or reasoning challenges while contributing to the same body of governed knowledge.

4.3 Educational Framework

Every diagnosis possesses an educational framework that defines what learners should understand after studying the diagnosis.

The educational framework includes:

- teaching objectives
- teaching rules
- clinical pearls
- clinical pitfalls
- recall prompts
- key concepts
- educational priorities

This framework defines the educational intent of the diagnosis before individual cases are written.

Cases should reinforce the educational framework rather than create it.

4.4 Diagnostic Knowledge Relationships

Clinical knowledge does not exist independently.

Diagnoses relate to one another through clinically meaningful relationships.

Examples include:

- differential diagnoses
- diagnostic mimics
- discriminating features
- supporting relationships
- prerequisite concepts
- shared mechanisms
- complication relationships

These relationships collectively form the diagnostic knowledge graph.

The knowledge graph is an editorial artifact requiring continuous governance rather than an automatically generated structure.

4.5 Educational Content

Educational content explains the diagnosis outside the context of gameplay.

It transforms diagnostic reasoning into structured learning.

Examples include:

- clinical summaries
- diagnostic reasoning explanations
- investigation guidance
- management principles
- examination findings
- pathophysiology
- complications
- references

Educational content should remain consistent with both clinical evidence and the educational framework.

4.6 Evidence

Evidence supports editorial confidence.

Evidence does not exist merely to satisfy citation requirements.

Its purpose is to justify educational decisions.

Evidence may support:

- clinical statements
- diagnostic relationships
- teaching rules
- management recommendations
- investigation guidance
- educational claims

The absence of supporting evidence does not necessarily imply incorrectness but always reduces editorial confidence.

4.7 Editorial Review

Editorial review transforms authored content into governed knowledge.

Review is a continuous process rather than a single approval event.

Editorial review evaluates:

- clinical correctness

- educational quality
- reasoning integrity
- internal consistency
- evidence sufficiency
- learner impact

Every review produces explicit editorial decisions that contribute to the governance history of an artifact.

4.8 Publication

Publication represents institutional confidence in an editorial artifact.

Publication is not merely a visibility state.

It signifies that the artifact has satisfied the required editorial standards for learner exposure.

Published knowledge remains subject to future review, improvement, and retirement.

5. Relationships Between Entities

The domain model intentionally separates educational concepts while recognising that they influence one another.

Examples include:

- Cases teach Diagnoses.
- Objectives define what Cases should teach.
- Teaching Rules reinforce Objectives.
- Clinical Pearls enrich understanding of Diagnoses.
- Differential Relationships distinguish Diagnoses.
- Evidence supports Educational Content.
- Reviews govern every artifact.
- Publication exposes governed knowledge to learners.

No artifact exists independently.

Every editorial artifact contributes to a larger network of diagnostic understanding.

6. Editorial Dependencies

Certain editorial entities depend upon others before meaningful work can begin.

For example:

A diagnosis requires metadata before educational generation.

Educational objectives should exist before comprehensive case authoring.

Teaching rules should exist before evaluating educational coverage.

Knowledge relationships should inform differential reasoning.

Cases should reinforce established educational intent.

Publication depends upon satisfactory governance of all required artifacts.

These dependencies define the logical progression of editorial work while remaining flexible enough to support iterative refinement.

7. Editorial Identity

Every editorial artifact possesses an independent identity.

An artifact may evolve through multiple revisions while remaining the same conceptual entity.

For example:

A clinical pearl may undergo numerous revisions.

A teaching objective may be rewritten.

A diagnostic relationship may gain stronger supporting evidence.

These changes improve the artifact without replacing its identity.

Editorial governance therefore focuses upon preserving the continuity of knowledge while allowing continuous refinement.

8. Artifact Independence

Editorial artifacts are intentionally modular.

A clinical pearl should not require rewriting a case.

Updating investigation guidance should not invalidate teaching objectives.

Improving a differential relationship should automatically strengthen future educational content without requiring manual duplication.

This modularity enables scalable editorial maintenance while preserving consistency throughout the knowledge ecosystem.

9. Domain Model Principles

The Editorial Domain Model is governed by the following principles.

Every diagnosis is the centre of a governed body of knowledge.

Cases are educational expressions rather than isolated documents.

Educational intent precedes educational delivery.

Knowledge relationships are first-class editorial artifacts.

Evidence strengthens editorial confidence.

Review governs all knowledge.

Publication represents institutional trust.

Knowledge remains continuously evolvable throughout its lifetime.

These principles define the conceptual structure upon which every subsequent component of the Wardle Editorial Operating System is built.

Closing Principle

The Editorial Domain Model describes what WEOS governs.

Subsequent sections describe how that knowledge is created, reviewed, improved, and entrusted to learners.

Part III — Editorial Capabilities

1. Purpose

The Editorial Domain Model defines what exists within the Wardle Editorial Operating System.

Editorial Capabilities define what the system must enable editors to accomplish.

Capabilities describe enduring editorial functions rather than software features.

They remain valid regardless of changes in user interface, implementation technology, or workflow design.

Every capability represents a core responsibility of WEOS in governing clinical knowledge.

No implementation should exist without supporting at least one editorial capability.

2. Capability Philosophy

WEOS is not organised around pages or administrative tools.

It is organised around editorial work.

Editors do not think in terms of viewing database entities.

Editors think in terms of completing meaningful editorial tasks.

Examples include:

- Create a diagnosis.
- Review a clinical case.
- Improve diagnostic reasoning.
- Approve a teaching objective.
- Validate a graph relationship.
- Publish educational knowledge.

Editorial capabilities therefore describe the work performed by clinicians rather than the software used to perform it.

3. Capability Categories

WEOS capabilities are organised into six major domains.

These domains collectively describe the complete lifecycle of clinical knowledge.

Knowledge Creation

↓

Knowledge Development

↓

Editorial Review

↓

Knowledge Governance

↓

Publication

↓

Continuous Improvement

Each domain contains multiple editorial capabilities.

4. Knowledge Creation

Knowledge creation establishes new editorial artifacts.

Creation may occur through human authoring, AI-assisted generation, or structured editorial workflows.

Knowledge creation does not imply publication.

Every newly created artifact enters editorial governance.

Core capabilities include:

- Diagnosis registration
- Metadata authoring
- Educational scaffold generation
- Clinical case authoring
- Educational content authoring
- Knowledge relationship creation
- Reference acquisition
- AI-assisted draft generation

The purpose of creation is to establish candidate knowledge for editorial evaluation.

5. Knowledge Development

Development improves existing knowledge.

Unlike creation, development assumes that an editorial artifact already exists.

Development capabilities include:

- Revising educational content
- Improving clinical reasoning
- Refining clue progression
- Expanding differential diagnoses
- Improving teaching objectives
- Updating teaching rules
- Strengthening clinical pearls
- Revising management guidance
- Expanding investigation guidance
- Improving educational consistency

Knowledge development is expected to continue throughout the lifetime of every artifact.

6. Editorial Review

Editorial review evaluates whether an artifact satisfies clinical and educational expectations.

Review does not create knowledge.

Review establishes confidence.

Review capabilities include:

- Clinical correctness review
- Educational quality review
- Diagnostic reasoning review
- Knowledge graph review
- Differential review
- Teaching objective review
- Teaching rule review
- Clinical pearl review
- Evidence review
- AI draft review
- Publication readiness review

Every review produces an explicit editorial decision.

Review never silently modifies knowledge.

7. Knowledge Governance

Governance maintains the integrity of the editorial ecosystem.

Governance differs from review.

Review evaluates individual artifacts.

Governance evaluates the relationships between artifacts.

Governance capabilities include:

- Detecting contradictory knowledge
- Identifying duplicate teaching
- Maintaining terminology consistency
- Managing knowledge relationships
- Validating curriculum coverage
- Monitoring evidence sufficiency
- Managing publication dependencies
- Maintaining editorial standards
- Recording governance decisions
- Managing lifecycle transitions

Governance ensures that independently reviewed artifacts collectively form a coherent educational system.

8. Publication

Publication transforms governed knowledge into learner-facing knowledge.

Publication capabilities include:

- Publication readiness assessment
- Dependency validation
- Final editorial approval
- Version publication
- Publication scheduling
- Learner release
- Publication withdrawal
- Publication supersession
- Publication history

Publication represents institutional trust rather than technical deployment.

9. Continuous Improvement

Clinical knowledge is expected to evolve.

Continuous improvement capabilities ensure that publication is never considered the end of editorial work.

These capabilities include:

- Evidence refresh
- Guideline monitoring
- Educational quality improvement
- Learner feedback integration
- Curriculum evolution
- Diagnostic graph refinement
- AI-assisted quality analysis
- Editorial analytics
- Knowledge retirement
- Knowledge replacement

Continuous improvement transforms WEOS from a publishing system into a living knowledge system.

10. Artificial Intelligence Capabilities

Artificial intelligence functions as an editorial assistant rather than an editorial authority.

AI capabilities include:

Knowledge Generation

Generating candidate educational artifacts.

Knowledge Analysis

Detecting inconsistencies, omissions, duplication, and opportunities for improvement.

Knowledge Recommendation

Suggesting educational improvements without modifying governed knowledge.

Knowledge Quality Assessment

Providing confidence estimates and quality projections for editorial consideration.

Editorial Assistance

Supporting editors during authoring, review, and governance activities.

AI capabilities always operate within human editorial governance.

AI contributes recommendations.

Editors make decisions.

11. Editorial Capability Dependencies

Capabilities are interconnected.

Certain capabilities depend upon the successful completion of others.

For example:

A clinical case cannot be meaningfully reviewed before it exists.

Publication cannot occur before governance.

Governance depends upon review.

Review depends upon authored knowledge.

Knowledge creation depends upon diagnosis registration.

These dependencies define the progression of editorial work while allowing iterative refinement where appropriate.

12. Capability Principles

Every editorial capability should satisfy the following principles.

Capability Independence

Capabilities should remain conceptually independent even when sharing implementation.

Capability Composability

Capabilities should combine to form larger editorial workflows without duplication.

Capability Explainability

Editors should understand why each capability exists.

Capability Auditability

Every meaningful editorial action should produce an auditable outcome.

Capability Extensibility

New capabilities should integrate naturally into the editorial operating model without requiring architectural redesign.

13. Capability Maturity

Capabilities mature over time.

Within WEOS, every capability progresses through four stages.

Foundational

The capability exists conceptually.

Operational

Editors can actively perform the capability.

Governed

The capability participates fully in editorial governance.

Intelligent

The capability proactively assists editors through recommendations, dependency analysis, and quality assessment.

The long-term objective of WEOS is not merely operational capability.

It is editorial intelligence.

14. The Capability Architecture

Editorial capabilities operate upon the Editorial Domain Model to continuously improve governed clinical knowledge.

Editorial Domain

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Editorial Capability

↓

Editorial Decision

↓

Governed Knowledge

↓

Learner Experience

↓

Editorial Feedback

↓

Knowledge Improvement

This progression represents the operational heart of WEOS.

Closing Principle

The purpose of WEOS is not to provide software features.

Its purpose is to enable the continuous creation, governance, publication, and improvement of trustworthy clinical knowledge.

Every implementation decision should strengthen one or more editorial capabilities.

Capabilities—not user interface components—define the enduring architecture of the Wardle Editorial Operating System.

Part IV — Knowledge Evolution & Editorial Lifecycles

1. Purpose

Clinical knowledge is not static.

Medical evidence evolves.

Educational priorities change.

Diagnostic reasoning improves.

Clinical guidelines are revised.

New misconceptions emerge.

The purpose of WEOS is therefore not simply to publish educational artifacts, but to continuously govern their evolution throughout their lifetime.

Every editorial artifact managed by WEOS progresses through a defined lifecycle.

These lifecycles establish how knowledge is created, improved, reviewed, trusted, maintained, and, when necessary, retired.

A lifecycle is not merely a workflow.

It represents the changing degree of editorial confidence placed in an artifact over time.

2. Knowledge Evolution Philosophy

Knowledge should never be considered complete.

Publication is not the conclusion of editorial work.

Rather, publication represents the point at which institutional confidence is sufficient for learner exposure.

After publication, knowledge remains subject to:

- evidence updates
- educational improvement
- curriculum changes
- reviewer feedback
- learner performance
- AI-assisted analysis
- clinical practice changes

WEOS therefore governs living knowledge rather than static publications.

3. Universal Editorial Lifecycle

Every governed editorial artifact follows the same conceptual progression.

Concept

↓

Draft

↓

Editorial Review

↓

Governance

↓

Approved

↓

Published

↓

Continuous Improvement

↓

Retired or Superseded

Not every artifact requires every stage.

However, every artifact progresses by increasing editorial confidence rather than increasing completeness.

4. Editorial Confidence

The purpose of a lifecycle is to represent confidence.

Each transition increases confidence through explicit editorial decisions.

Confidence is established by evaluating:

- clinical correctness
- educational quality
- evidence sufficiency
- internal consistency
- curriculum suitability
- governance requirements

Confidence is never inferred from artifact age or publication history.

It is earned through review.

5. Diagnosis Lifecycle

The diagnosis is the highest-level governed entity within WEOS.

Its lifecycle coordinates every subordinate editorial artifact.

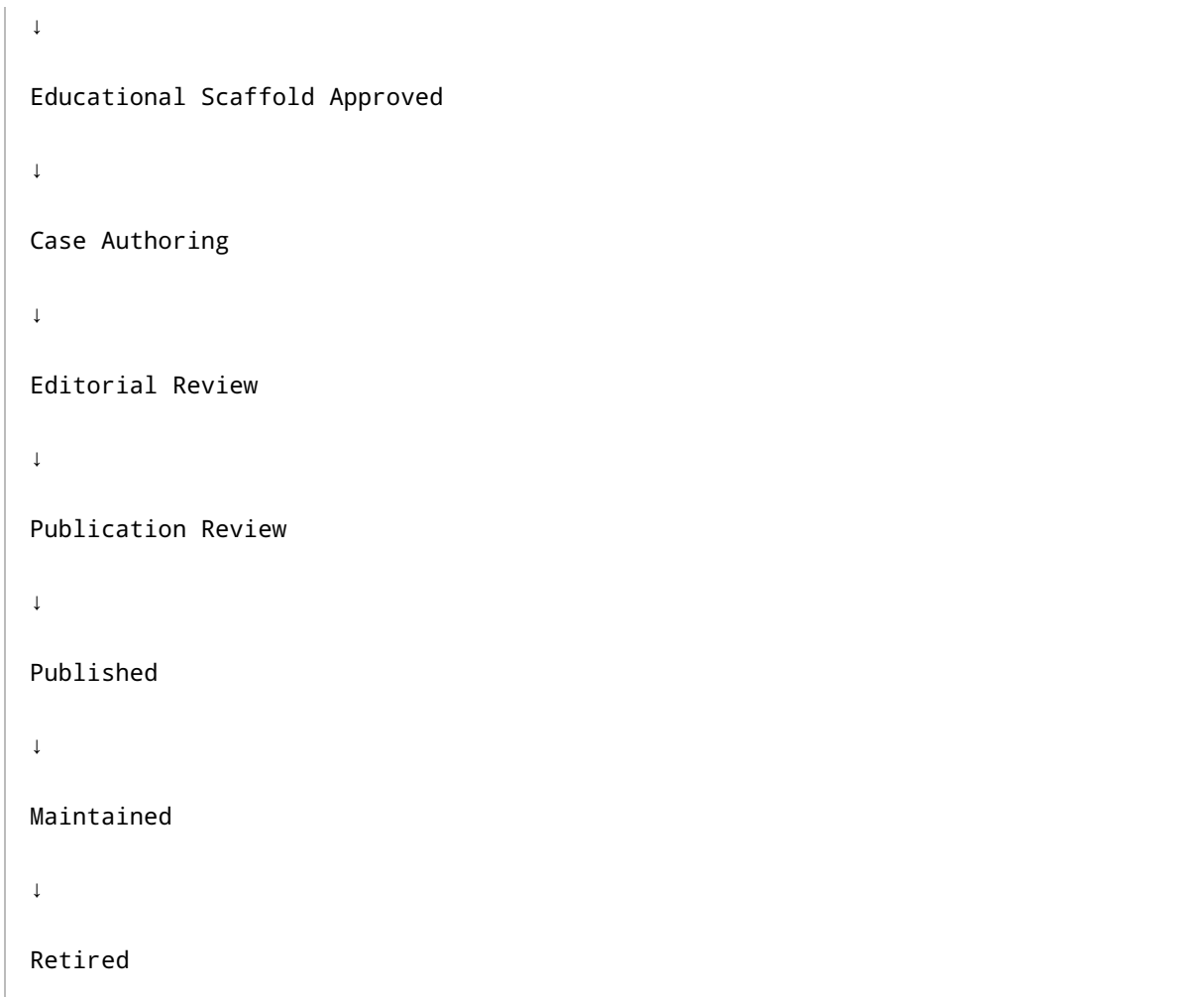
Registry Entry

↓

Metadata Complete

↓

Educational Scaffold Required



A diagnosis may remain in any stage indefinitely until sufficient editorial confidence is achieved.

Publication never bypasses governance.

6. Educational Scaffold Lifecycle

Before comprehensive educational content can exist, every diagnosis requires an educational scaffold.

The scaffold defines the educational intent of the diagnosis.

The scaffold typically includes:

- learning objectives
- teaching rules
- differential framework
- diagnostic discriminators

- educational priorities
- clinical pearls
- pitfalls

Lifecycle:

Not Started

↓

AI or Human Draft

↓

Editorial Review

↓

Approved

↓

Available for Case Development

↓

Maintained

The scaffold precedes large-scale case generation because it establishes what future cases are intended to teach.

7. Clinical Case Lifecycle

Clinical cases represent educational implementations of the scaffold.

Their lifecycle reflects progressive refinement of diagnostic reasoning.

Draft

↓

Clue Review



A case may remain approved without being published if broader diagnosis-level publication requirements have not yet been satisfied.

8. Knowledge Relationship Lifecycle

Knowledge relationships evolve independently of cases.

Relationships include:

- differential diagnoses
- discriminators
- supporting relationships
- prerequisite concepts
- related diagnoses

Lifecycle:

Suggested

↓

Evidence Assessment

↓

Editorial Review

↓

Approved

↓

Knowledge Graph

↓

Continuous Validation

Relationships may be strengthened, weakened, or removed as evidence evolves.

9. Educational Content Lifecycle

Educational content supports the diagnosis outside gameplay.

Lifecycle:

Draft

↓

Review

↓

Approved

↓

Published

↓

Evidence Refresh

↓

Revision

↓

Republished

Educational revisions should preserve historical accountability while continuously improving learner understanding.

10. AI Draft Lifecycle

Artificial intelligence generates candidate knowledge rather than governed knowledge.

AI-generated artifacts never bypass editorial review.

Lifecycle:

Generation

↓

Pending Human Review

↓

Approved

or

Rejected

or

Returned for Revision

↓

Governed Artifact

Approval transforms an AI draft into an editorial artifact.

The draft itself remains part of the editorial history.

11. Editorial Review Lifecycle

Review itself is a governed process.

Lifecycle:

Review Requested

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In Progress

↓

Evidence Requested

↓

Decision Recorded

↓

Closed

↓

Reopened (if necessary)

Editorial reviews are historical records rather than transient activities.

12. Publication Lifecycle

Publication represents institutional trust.

Lifecycle:

Publication Assessment

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Publication Approved

↓

Scheduled

↓

Released

↓

Maintained

↓

Withdrawn

or

Superseded

Publication status should never replace editorial status.

An artifact may remain editorially approved while awaiting publication.

13. Continuous Maintenance

Maintenance is an explicit lifecycle stage.

Published knowledge should be continuously monitored for:

- outdated evidence
- inconsistent terminology
- educational weaknesses
- learner misconceptions
- duplicated teaching
- missing differentials

- evolving clinical guidance

Maintenance transforms publication into an ongoing editorial responsibility.

14. Retirement

Not all knowledge remains educationally valuable forever.

Artifacts may eventually be retired due to:

- replacement
- obsolescence
- curriculum redesign
- merged diagnoses
- deprecated clinical practice
- editorial restructuring

Retirement preserves historical accountability while preventing obsolete knowledge from influencing future learners.

15. Lifecycle Independence

Every editorial artifact possesses its own lifecycle.

A diagnosis may be published while new cases remain under review.

A teaching objective may be revised without affecting approved graph relationships.

A clinical pearl may be updated independently of case revisions.

This independence enables continuous improvement without requiring wholesale republication.

16. Lifecycle Dependencies

Although lifecycles are independent, they remain logically connected.

Examples include:

- Educational scaffolds support case development.
- Cases reinforce learning objectives.
- Knowledge relationships strengthen differential reasoning.

- Educational content explains published knowledge.
- Evidence increases editorial confidence.
- Publication depends upon sufficient governance across all required artifacts.

Dependencies coordinate editorial progression without preventing iterative improvement.

17. Lifecycle Principles

The lifecycle architecture of WEOS is governed by the following principles.

Every artifact has a lifecycle.

Every transition represents increased editorial confidence.

Publication is a governance milestone rather than a terminal state.

Maintenance is a permanent editorial responsibility.

Knowledge remains continuously evolvable.

Editorial history is preserved.

Independent artifacts evolve independently.

Dependencies guide progression without restricting improvement.

Closing Principle

The purpose of editorial lifecycles is not to move content through software states.

Their purpose is to govern the continuous evolution of trustworthy clinical knowledge.

Every transition within WEOS should represent a deliberate increase in institutional confidence, ensuring that medical knowledge becomes progressively more reliable, more educationally valuable, and more clinically defensible over time.

Part V — Knowledge Production & Editorial Intelligence

1. Purpose

The purpose of the Wardle Editorial Operating System is not simply to govern existing knowledge.

It must also provide a structured, transparent, and clinically safe mechanism through which new knowledge enters the editorial ecosystem.

Knowledge production encompasses every process responsible for introducing new editorial artifacts into WEOS.

These artifacts may originate from human authors, artificial intelligence, imported clinical guidance, existing editorial content, or future knowledge sources.

Regardless of origin, no artifact enters the governed knowledge base directly.

Every artifact begins life as candidate knowledge.

Only after appropriate editorial governance does candidate knowledge become trusted institutional knowledge.

2. Philosophy of Knowledge Production

Clinical knowledge should never appear spontaneously within the editorial system.

Every editorial artifact must have a clearly understood origin, purpose, and intended educational contribution.

The source of knowledge influences editorial confidence but never replaces editorial governance.

A diagnosis written by an experienced clinician, an AI-generated teaching objective, or an imported clinical guideline all enter WEOS through the same philosophical principle:

Knowledge is proposed before it is trusted.

Editorial confidence is earned through governance rather than inherited from the source.

3. Editorial Intelligence

Editorial Intelligence is the decision-making layer responsible for understanding the editorial state of the knowledge ecosystem and determining what work should occur next.

Editorial Intelligence is distinct from Artificial Intelligence.

Artificial Intelligence refers to computational models capable of generating or analysing information.

Editorial Intelligence refers to the governance framework that evaluates editorial completeness, detects deficiencies, orchestrates production workflows, and recommends editorial actions.

Artificial Intelligence may contribute to Editorial Intelligence, but Editorial Intelligence remains a broader architectural concept that includes deterministic rules, editorial policies, governance logic, quality models, dependency analysis, and human judgment.

Editorial Intelligence exists to answer questions such as:

- What knowledge is missing?
- Which diagnoses require development?
- Which educational objectives remain uncovered?
- Which cases should be generated next?
- Which graph relationships require review?
- Which artifacts have become outdated?
- Which published knowledge requires maintenance?

The objective of Editorial Intelligence is not automation.

Its objective is editorial awareness.

4. Knowledge Producers

Knowledge may originate from multiple producers.

WEOS recognises the origin of knowledge independently from its editorial status.

Knowledge producers include, but are not limited to:

- Human authors
- Artificial Intelligence
- Clinical guideline imports
- Editorial migrations
- Knowledge graph extraction
- Curriculum imports

- Evidence extraction systems
- Future automated editorial systems

Every artifact should preserve its production history.

The producer of an artifact contributes provenance, not authority.

Authority is established through editorial governance.

5. Candidate Knowledge

Before becoming governed knowledge, every editorial artifact exists as Candidate Knowledge.

Candidate Knowledge represents proposed educational material awaiting editorial evaluation.

Examples include:

- proposed teaching objectives
- proposed teaching rules
- proposed clinical pearls
- proposed pitfalls
- proposed diagnostic relationships
- proposed educational summaries
- proposed clinical cases
- proposed recall prompts
- proposed investigation guidance
- proposed management guidance

Candidate Knowledge is intentionally separated from governed editorial content.

This separation protects learners from unreviewed educational material while allowing extensive experimentation during editorial development.

6. Draft Artifacts

Candidate Knowledge is represented through Draft Artifacts.

Drafts are temporary editorial objects that exist solely to facilitate review.

Drafts are not educational resources.

They are editorial proposals.

Draft artifacts may include:

- Diagnosis Draft
- Educational Scaffold Draft
- Teaching Objective Draft
- Teaching Rule Draft
- Clinical Pearl Draft
- Pitfall Draft
- Differential Draft
- Knowledge Relationship Draft
- Education Section Draft
- Clinical Case Draft
- Clue Draft
- Explanation Draft
- Recall Prompt Draft

Every draft possesses:

- editorial identity
- producer
- generation history
- editorial rationale
- review status
- revision history
- governance history

Approval transforms a draft into a governed editorial artifact while preserving its editorial history.

7. Educational Scaffold Generation

Educational development begins with the educational scaffold.

The scaffold defines the educational intent of a diagnosis before comprehensive content development begins.

The scaffold commonly establishes:

- educational objectives
- teaching priorities
- diagnostic discriminators
- differential framework
- clinical pearls
- pitfalls
- recall priorities
- expected learner outcomes

Cases should emerge from the scaffold.

They should not define it.

This principle ensures that every case contributes intentionally to diagnosis-level educational goals.

8. Dependency-Driven Generation

Knowledge production within WEOS is governed by editorial dependencies rather than arbitrary generation requests.

Generation occurs because prerequisite editorial structures indicate that new work is required.

For example:

A diagnosis with complete metadata but no educational scaffold should recommend scaffold generation.

A diagnosis with an approved scaffold but no cases should recommend case generation.

A diagnosis with cases but incomplete educational explanations should recommend education generation.

A diagnosis with established reasoning but missing graph relationships should recommend relationship generation.

Generation therefore responds to editorial needs rather than isolated user commands.

9. Editorial Suggestions

Not every recommendation should immediately produce new knowledge.

Editorial Intelligence frequently produces Suggestions rather than Drafts.

Suggestions identify opportunities for improvement without creating candidate artifacts.

Examples include:

- suggested teaching objective
- suggested differential diagnosis
- suggested graph relationship
- suggested clue refinement
- suggested investigation
- suggested management update

- suggested educational gap
- suggested duplicate teaching
- suggested unsupported claim review

Suggestions represent editorial opportunities.

Editors decide whether those opportunities should become candidate knowledge.

10. Generation Policies

Each editorial artifact follows an explicit generation policy.

Generation policies define:

- whether AI may generate the artifact
- whether manual authoring is supported
- whether regeneration is permitted
- whether evidence is required before review
- which reviewer role is required
- which governance pathway applies

Generation policy is independent of implementation.

It reflects institutional editorial standards rather than technical capability.

11. Artificial Intelligence Within WEOS

Artificial Intelligence functions as an editorial contributor.

It performs work that assists human editors without replacing editorial judgment.

AI may contribute by:

- drafting educational content
- generating clinical cases
- proposing teaching objectives
- suggesting graph relationships
- identifying educational gaps
- detecting duplicated teaching
- analysing reasoning quality
- recommending curriculum improvements
- identifying inconsistencies
- estimating editorial confidence

Artificial Intelligence never establishes institutional trust.

Editorial trust is created exclusively through governed human decisions.

12. Human Editorial Responsibility

Clinical knowledge ultimately remains a human responsibility.

Editors determine:

- clinical correctness
- educational value
- evidence sufficiency
- publication suitability
- learner safety
- institutional confidence

Artificial Intelligence accelerates editorial work.

Editors govern editorial knowledge.

This distinction remains fundamental to every future capability within WEOS.

13. Knowledge Production Lifecycle

Knowledge production follows a universal progression.

Editorial Need Identified

↓

Generation Recommended

↓

Candidate Knowledge Created

↓

Editorial Review

↓

Governance

↓

Approved Editorial Artifact

↓

Publication Eligibility

↓

Published Knowledge

This lifecycle applies regardless of the producer.

14. Knowledge Provenance

Every editorial artifact should preserve complete provenance.

Provenance establishes transparency throughout the editorial lifecycle.

The provenance of an artifact should include:

- origin
- producer
- generation method
- creation time
- supporting context
- review history
- governance decisions
- publication history

Provenance enables explainability without implying correctness.

An artifact is trusted because it has been governed, not because of where it originated.

15. Principles of Knowledge Production

Knowledge production within WEOS is governed by the following principles.

Every editorial artifact begins as candidate knowledge.

Knowledge origin never replaces editorial governance.

Editorial Intelligence orchestrates production.

Artificial Intelligence assists editorial work but does not govern it.

Educational scaffolds precede comprehensive content generation.

Generation should follow editorial dependencies.

Editorial suggestions identify work before creating knowledge.

Every artifact preserves provenance.

Editorial trust is earned through review.

Institutional confidence emerges from governance rather than automation.

Closing Principle

The purpose of Knowledge Production is not to generate more educational content.

Its purpose is to introduce clinically meaningful, educationally intentional, and transparently governed candidate knowledge into the editorial ecosystem.

Within WEOS, knowledge is not trusted because it has been created.

Knowledge is trusted because it has been governed.

Part VI — Editorial Intelligence & Decision Architecture

1. Purpose

The purpose of the Wardle Editorial Operating System is not merely to store, generate, or review clinical knowledge.

It must also interpret the editorial state of that knowledge.

Editorial Intelligence is the reasoning layer that transforms governed information into meaningful editorial decisions.

Rather than presenting editors with isolated facts, Editorial Intelligence synthesises knowledge across the editorial ecosystem to answer practical questions such as:

- What requires attention?
- What is safe to approve?
- What prevents publication?
- What should be generated next?
- Which diagnosis is incomplete?
- Which evidence is insufficient?
- Which educational objectives remain uncovered?

Editorial Intelligence exists to reduce editorial complexity without reducing editorial transparency.

2. Editorial Intelligence

Editorial Intelligence is the governed reasoning layer of WEOS.

It continuously interprets editorial knowledge, evaluates dependencies, applies governance policies, and produces recommendations that assist human decision-making.

Editorial Intelligence does not replace editors.

It supports editors by transforming fragmented editorial information into coherent understanding.

Editorial Intelligence combines:

- editorial policies

- lifecycle state
- dependency analysis
- evidence assessment
- governance rules
- quality models
- knowledge relationships
- AI recommendations
- human decisions

to produce meaningful editorial guidance.

3. Editorial Questions

Every capability within WEOS ultimately exists to answer an editorial question.

Examples include:

- Is this diagnosis ready for scaffold generation?
- Is this teaching objective clinically appropriate?
- Is this graph relationship defensible?
- Is this case educationally safe?
- Is learner exposure appropriate?
- Does this diagnosis require another case?
- Which evidence remains unsupported?
- Which differentials are insufficiently represented?

Editorial Intelligence exists to answer these questions consistently throughout the system.

4. From Data to Decisions

WEOS distinguishes between information and decisions.

Information describes the editorial state.

Decisions interpret that state.

For example:

Information may indicate:

- three unsupported claims
- one unresolved differential
- incomplete educational objectives

- pending AI review

Editorial Intelligence transforms this information into a decision.

For example:

"This diagnosis is not yet ready for publication because unresolved educational and evidence deficiencies remain."

The purpose of Editorial Intelligence is therefore interpretation rather than presentation.

5. Editorial Assessments

Editorial Intelligence evaluates artifacts through structured assessments.

An assessment represents an editorial interpretation rather than a stored property.

Examples include:

- Clinical Assessment
- Educational Assessment
- Evidence Assessment
- Reasoning Assessment
- Knowledge Relationship Assessment
- Curriculum Assessment
- Publication Assessment

Assessments remain explainable.

Editors should always understand why an assessment reached its conclusion.

6. Editorial Decisions

Every governed artifact eventually reaches an editorial decision.

Editorial decisions are explicit, auditable outcomes produced through governance.

Typical decisions include:

- Approve
- Reject
- Request Revision
- Request Evidence

- Recommend Regeneration
- Defer
- Publish
- Withdraw
- Supersede

Every decision records:

- rationale
- supporting assessments
- responsible reviewer
- affected artifacts
- governance history

Editorial decisions are institutional decisions rather than user interface events.

7. Editorial Recommendations

Editorial Intelligence may recommend work without making editorial decisions.

Recommendations assist editors by identifying opportunities for improvement.

Examples include:

- generate another case
- strengthen differential coverage
- revise clue progression
- expand investigation guidance
- review unsupported claims
- add graph relationships
- improve recall prompts
- update references

Recommendations guide editorial effort while preserving human authority.

8. Editorial Readiness

Readiness is a computed interpretation of editorial maturity.

Readiness is never manually assigned.

Instead, Editorial Intelligence continuously evaluates whether an artifact satisfies the requirements for progressing to its next lifecycle stage.

Examples include:

Generation Readiness

Editorial Review Readiness

Publication Readiness

Maintenance Readiness

Retirement Readiness

Readiness should always explain itself.

Editors should understand:

- why an artifact is ready
- why it is not ready
- what remains outstanding
- what action should occur next

9. Editorial Dependencies

Knowledge within WEOS is interconnected.

Editorial Intelligence continuously evaluates dependencies between artifacts.

Examples include:

Educational objectives supporting clinical cases.

Teaching rules supporting learning outcomes.

Graph relationships strengthening differential reasoning.

Evidence supporting educational content.

Cases reinforcing curriculum coverage.

Publication depending upon governance completeness.

Dependency analysis ensures that editorial progression occurs logically while remaining flexible enough to support iterative improvement.

10. Editorial Confidence

Editorial confidence represents the degree of institutional trust placed in an artifact.

Confidence is not equivalent to certainty.

Instead, it reflects the cumulative outcome of editorial governance.

Confidence is strengthened through:

- review
- evidence
- consistency
- educational quality
- reasoning integrity
- governance history

Confidence decreases when:

- evidence becomes outdated
- inconsistencies emerge
- educational weaknesses are identified
- clinical guidance changes
- governance deficiencies remain unresolved

Editorial confidence continuously evolves throughout an artifact's lifecycle.

11. Editorial Explainability

Every recommendation, assessment, and decision produced by Editorial Intelligence should be explainable.

Editors should never encounter unexplained conclusions.

Every editorial conclusion should answer:

- What was evaluated?
- Which policies were applied?
- Which evidence influenced the result?
- Which dependencies contributed?
- Which recommendations were generated?
- What actions remain available?

Explainability strengthens institutional trust by making governance transparent.

12. Editorial Decision Patterns

Although different artifacts undergo different review processes, every editorial decision follows the same conceptual pattern.

Editorial Artifact

↓

Current Editorial State

↓

Editorial Assessments

↓

Dependency Analysis

↓

Evidence Evaluation

↓

Governance Policies

↓

Editorial Intelligence

↓

Recommendation

↓

Human Editorial Decision

↓

Governed Outcome

This pattern provides consistency across every editorial capability within WEOS.

13. Editorial Work Prioritisation

Editorial Intelligence continuously evaluates the editorial ecosystem to determine where effort should be directed.

Prioritisation considers:

- learner impact
- publication blockers
- missing educational coverage
- unresolved governance issues
- evidence deficiencies
- incomplete diagnoses
- curriculum importance
- maintenance requirements

Prioritisation assists editors without replacing editorial judgement.

14. Editorial Intelligence Principles

Editorial Intelligence operates according to the following principles.

Interpretation is more valuable than presentation.

Recommendations should reduce editorial complexity.

Decisions should always be explainable.

Readiness is computed rather than assigned.

Dependencies should guide editorial progression.

Recommendations assist but never replace governance.

Human editors remain responsible for institutional trust.

Editorial Intelligence exists to improve judgement rather than automate it.

15. Relationship to Artificial Intelligence

Artificial Intelligence contributes information.

Editorial Intelligence governs decisions.

Artificial Intelligence may identify missing differentials.

Editorial Intelligence determines whether those differentials should enter editorial review.

Artificial Intelligence may generate a clinical pearl.

Editorial Intelligence determines whether the draft satisfies generation policies and requires governance.

This separation ensures that the architecture remains stable regardless of future advances in artificial intelligence.

Editorial Intelligence is therefore an enduring architectural capability rather than a specific technological implementation.

Closing Principle

Editorial Intelligence is the reasoning engine of the Wardle Editorial Operating System.

Its purpose is not to replace human editors, but to continuously interpret the evolving state of clinical knowledge, reveal what matters, explain why it matters, and support consistent, transparent, and defensible editorial decisions.

Within WEOS, information becomes knowledge through governance.

Knowledge becomes trusted through editorial decisions.

Editorial Intelligence exists to make those decisions understandable, consistent, and scalable.

Part VII — Editorial Workspace Architecture

1. Purpose

The Editorial Workspace is the primary environment through which editors interact with the Wardle Editorial Operating System (WEOS).

Its purpose is not to expose stored information.

Its purpose is to support editorial thinking.

The workspace should continuously assist editors in understanding the current state of governed clinical knowledge, identifying what requires attention, completing editorial work, and making safe, explainable decisions.

The workspace therefore exists to support editorial judgement rather than administrative navigation.

2. Workspace Philosophy

The Editorial Workspace is designed around editorial work rather than data structures.

Editors should never need to understand how information is stored internally.

Instead, the workspace should present knowledge according to the editorial questions being answered.

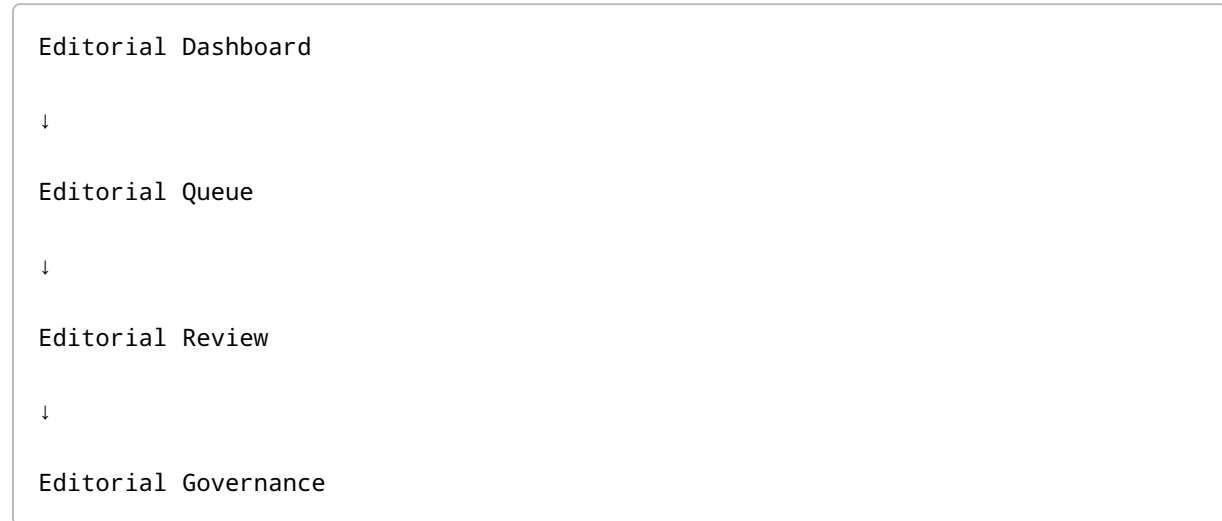
Examples include:

- What requires my attention?
- Is this diagnosis educationally complete?
- What should be generated next?
- Can this artifact be approved?
- What prevents publication?
- Which knowledge requires maintenance?

The organisation of the workspace should therefore reflect editorial reasoning rather than database organisation.

3. Workspace Model

The Editorial Workspace consists of four complementary environments.



Each environment serves a distinct purpose while sharing the same governed knowledge model.

Editors move between environments as editorial work progresses.

4. Editorial Dashboard

The Dashboard provides situational awareness.

Its purpose is to answer:

"What is the current state of the editorial ecosystem?"

The Dashboard should summarise:

- diagnosis completeness
- editorial readiness
- publication readiness
- governance status
- educational coverage
- outstanding work
- maintenance requirements

The Dashboard is observational.

It supports understanding rather than decision-making.

5. Editorial Queue

The Queue represents work awaiting editorial attention.

Unlike the Dashboard, the Queue is action-oriented.

The Queue should organise work according to editorial priorities rather than artifact type.

Examples include:

- diagnoses requiring scaffolding
- AI drafts awaiting review
- cases awaiting approval
- graph relationships requiring governance
- unsupported claims
- publication blockers
- evidence updates
- maintenance tasks

The Queue answers:

"What should I work on next?"

6. Editorial Review

Editorial Review is where candidate knowledge becomes governed knowledge.

Review is centred upon a single editorial task.

Examples include:

- reviewing a case
- reviewing a teaching objective
- reviewing a graph relationship
- reviewing educational content
- reviewing a clinical pearl

Every review should provide the complete context required for safe editorial decision-making.

Editors should not need to navigate between multiple locations to understand an artifact.

7. Editorial Governance

Governance evaluates the editorial system rather than individual artifacts.

Governance focuses upon:

- consistency
- completeness
- evidence
- relationships
- curriculum alignment
- editorial quality
- publication standards

Governance provides confidence that independently reviewed artifacts collectively form a coherent educational system.

8. Editorial Work Packets

Editorial work is performed through Work Packets.

A Work Packet is the smallest complete unit of editorial work.

A packet contains everything required to complete one editorial decision.

Examples include:

- Case Review Packet
- Teaching Rule Review Packet
- Clinical Pearl Review Packet
- Differential Review Packet
- Knowledge Relationship Review Packet
- Educational Section Review Packet
- Publication Packet

A Work Packet should answer:

- What is being reviewed?
- Why does it exist?
- What supports it?
- What remains unresolved?
- Which decision is required?
- What happens next?

Editors should complete work from within the packet without unnecessary context switching.

9. Workspace Layers

The Editorial Workspace presents information through progressively deeper layers.

Editorial Awareness

↓

Editorial Work

↓

Editorial Decision

↓

Editorial Detail

↓

Editorial History

Editors should begin with broad situational awareness and progressively reveal detail only as required.

The workspace should minimise cognitive load while preserving transparency.

10. Progressive Disclosure

Not every editorial detail deserves equal prominence.

The workspace should progressively reveal information according to editorial importance.

High-level decisions should appear before implementation detail.

Examples include:

- publication recommendation before validation logs
- governance blockers before audit metadata
- educational deficiencies before database identifiers
- evidence status before raw references

Editors should first understand what requires action before investigating why.

11. Editorial Context

Every editorial task should preserve context.

Editors should always understand:

- the diagnosis
- the educational purpose
- related artifacts
- lifecycle stage
- dependencies
- previous decisions
- current recommendations

Context should accompany editorial work rather than requiring repeated navigation.

12. Workspace Navigation

Navigation exists to support editorial thinking rather than expose application structure.

Movement throughout the workspace should follow editorial progression.

Examples include:

Diagnosis



Educational Scaffold



Clinical Case



Evidence



Review



Publication

rather than arbitrary navigation between unrelated administrative pages.

13. Workspace Consistency

Every editorial surface should present information using a consistent conceptual structure.

Editors should encounter familiar patterns regardless of artifact type.

Every review surface should communicate:

- identity
- editorial purpose
- current state
- supporting evidence
- dependencies
- recommendations
- available decisions
- governance history

Consistency reduces cognitive effort and improves editorial confidence.

14. Workspace Intelligence

The Editorial Workspace should continuously expose the output of Editorial Intelligence.

Rather than merely displaying stored information, the workspace should communicate:

- recommended work
- publication readiness
- generation readiness
- governance concerns
- educational gaps
- dependency conflicts
- maintenance priorities

Editors should spend their effort making decisions rather than searching for information.

15. Workspace Principles

The Editorial Workspace operates according to the following principles.

Editorial work should determine workspace organisation.

Context should remain visible throughout editorial tasks.

Work Packets should contain complete decision context.

Navigation should follow editorial progression.

Progressive disclosure should reduce cognitive load.

Recommendations should precede raw information.

Governance should remain transparent.

Editorial Intelligence should guide attention without replacing judgement.

Every editorial decision should be explainable.

The workspace should make editorial work easier, not merely possible.

Closing Principle

The Editorial Workspace is not an administrative interface.

It is the operational environment through which governed clinical knowledge is created, evaluated, maintained, and entrusted to learners.

Its success is measured not by the quantity of information displayed, but by its ability to help editors make safe, consistent, efficient, and clinically defensible decisions.

Wardle Editorial Operating System (WEOS)

Part VIII - Institutional Editorial Governance

WEOS-ARCH-008 | Version 0.1 | Draft for approval

1. Purpose

The Wardle Editorial Operating System governs clinical knowledge on behalf of learners, educators, reviewers, and institutions.

Clinical knowledge does not become trustworthy merely because it has been written, generated, reviewed, or technically published.

It becomes institutionally trusted when qualified people exercise defined authority through transparent, accountable, and reproducible editorial governance.

The purpose of Institutional Editorial Governance is to define how responsibility, authority, accountability, and oversight operate throughout WEOS.

This chapter establishes:

- who contributes editorial knowledge;
- who may evaluate it;
- who may make editorial decisions;
- how authority is assigned;
- how disagreements are resolved;
- how conflicts of interest are managed;
- how evidence and high-risk content receive appropriate oversight;
- how publication authority is exercised;
- how governance remains accountable;
- how editorial governance scales from individual editors to institutional and multi-institutional structures.

Institutional Editorial Governance does not define user-interface permissions or specific organisational job titles.

It defines the enduring authority model that every implementation and editorial organisation must preserve.

2. Philosophy of Institutional Governance

WEOS treats editorial governance as an institutional responsibility rather than an individual activity.

An individual editor may possess expertise.

A reviewer may possess experience.

A specialty board may possess collective authority.

However, expertise alone does not establish institutional trust.

Institutional trust emerges when expertise is exercised through:

- defined responsibilities;
- appropriate scope of authority;
- transparent assessment;
- explicit decisions;
- preserved rationale;
- evidence-informed judgement;
- accountable publication;
- continuing oversight.

Governance should protect the editorial system from:

- unreviewed clinical assertions;
- informal approval;
- authority based solely on system access;
- publication without accountability;
- concentration of incompatible responsibilities;
- hidden conflicts of interest;
- inconsistent specialty standards;
- unresolved reviewer disagreement;
- unsupported exceptions;
- loss of institutional memory.

Institutional governance strengthens clinical judgement by placing it within a durable framework of responsibility.

3. Institutional Trust

Institutional trust represents the confidence that an organisation places in governed clinical knowledge.

It is not equivalent to confidence in a particular author, editor, artificial-intelligence system, or external source.

Institutional trust is created through the combined effect of:

- stable editorial identity;
- appropriate clinical scope;
- evidence;
- qualified review;
- educational evaluation;
- governance decisions;
- dependency assessment;

- publication authority;
- maintenance;
- preserved history.

Institutional trust belongs to the governed artifact and its approved version.

It does not transfer automatically from the reputation of its creator.

Institutional trust is earned through governance rather than inherited from origin.

4. Governance Scope

Institutional Editorial Governance operates across complementary levels.

4.1 Artifact Governance

Artifact Governance evaluates an individual editorial artifact.

It asks whether the artifact is clinically defensible, educationally appropriate, evidence-supported, correctly scoped, and eligible to progress.

4.2 Diagnosis Governance

Diagnosis Governance evaluates the complete editorial foundation of a diagnosis, including identity, metadata, educational intent, cases, education, relationships, evidence, dependencies, and maintenance.

4.3 Specialty Governance

Specialty Governance evaluates knowledge within the clinical standards and educational priorities of a specialty or subspecialty.

4.4 Curriculum Governance

Curriculum Governance evaluates whether independently governed artifacts collectively form an appropriate programme of learning.

4.5 Institutional Governance

Institutional Governance oversees authority, policies, quality, evidence, publication controls, maintenance, auditability, and organisational accountability.

An artifact may be clinically acceptable while its diagnosis remains incomplete.

A diagnosis may be well developed while specialty-level inconsistencies remain unresolved.

Institutional governance ensures that trust is evaluated at every relevant level.

5. Editorial Authority

Editorial authority is the recognised ability to make a governed decision within a defined scope.

Authority should reflect:

- professional competence;
- editorial responsibility;
- artifact risk;
- decision consequence;
- clinical scope;
- educational scope;
- institutional mandate.

Authority must not be inferred solely from technical access.

The ability to change a database record, apply a migration, or operate an administrative interface does not confer clinical or publication authority.

Editorial authority should always answer:

- Who may make this decision?
- Which artifact or domain does the authority cover?
- Which conditions limit the authority?
- Which decisions require additional review?
- Who remains accountable for the outcome?
- How is the decision recorded?

6. Editorial Roles

WEOS recognises editorial roles according to responsibility rather than software permission.

6.1 Contributor

A Contributor introduces candidate knowledge, identified errors, references, feedback, or maintenance concerns.

Contribution establishes provenance but not authority.

6.2 Editorial Author

An Editorial Author develops candidate artifacts according to approved educational intent and responds to review findings.

Authorship does not automatically confer approval authority.

6.3 Clinical Reviewer

A Clinical Reviewer evaluates factual correctness, diagnostic interpretation, investigations, management, safety, severity, population applicability, and risk of harmful misunderstanding.

6.4 Educational Reviewer

An Educational Reviewer evaluates learning purpose, reasoning progression, clue sequencing, learner level, misconception prevention, educational usefulness, and curriculum alignment.

6.5 Knowledge Governance Reviewer

A Knowledge Governance Reviewer evaluates structured relationships, graph facts, reasoning paths, differential mappings, contradiction, duplication, provenance, and downstream consequences.

6.6 Evidence Reviewer

An Evidence Reviewer evaluates source authority, recency, applicability, methodological quality, claim alignment, and conflicting evidence.

6.7 Specialty Reviewer

A Specialty Reviewer provides deeper authority for specialist diagnostic nuance, advanced investigations, high-risk management, uncommon disease, or evolving guidance.

6.8 Senior Editor

A Senior Editor resolves complex decisions, coordinates multidisciplinary review, supervises specialty governance, and authorises high-impact editorial outcomes within their mandate.

6.9 Publication Authority

The Publication Authority determines whether governed knowledge may become learner-facing or institutionally operational.

6.10 Editorial Operations Lead

The Editorial Operations Lead assigns work, monitors queues, coordinates reviewers, tracks maintenance, and escalates operational risk without silently exercising clinical authority.

6.11 System Administrator

A System Administrator manages access, migrations, restoration, and technical operation.

Administrative capability does not create editorial authority.

7. Role Qualification and Scope

Editorial authority should be based upon demonstrable suitability for the decision being made.

Suitability may include:

- clinical training;
- specialty experience;
- educational experience;
- editorial competence;
- familiarity with WEOS standards;

- evidence-appraisal competence;
- knowledge-governance competence;
- institutional appointment;
- continuing participation in quality review.

Qualification should be proportionate to risk.

Where an artifact requires several forms of authority, governance should combine them rather than assume one reviewer possesses every necessary perspective.

8. Editorial Ownership

Every actively governed artifact should have identifiable editorial ownership.

Ownership establishes responsibility for continued stewardship rather than unrestricted decision authority.

Ownership may exist at artifact, diagnosis, specialty, curriculum, publication, and maintenance levels.

Ownership should answer:

- Who ensures work progresses?
- Who monitors unresolved concerns?
- Who coordinates required reviewers?
- Who responds when evidence changes?
- Who ensures published knowledge remains maintained?

Ownership must remain transferable without losing governance history.

9. Assignment of Editorial Work

Editorial work should be assigned according to competence, artifact type, specialty, risk, learner impact, urgency, independence, and availability.

Assignment creates responsibility to review or act.

It does not predetermine the decision.

Reviewers should remain free to approve, reject, request revision, request evidence, defer, escalate, declare insufficient expertise, or declare a conflict of interest.

10. Separation of Duties

Certain responsibilities should remain separate where combining them would weaken governance.

Separation may require that:

- an author does not provide the only approval of their own work;
- technical application does not substitute for editorial approval;
- publication is authorised separately from generation;
- high-risk clinical content receives independent review;
- disputed decisions are adjudicated independently;
- emergency actions receive retrospective review;
- migrations affecting governed knowledge are authorised before execution.

Where small teams require overlapping roles, governance should compensate through disclosure, additional review, preserved rationale, audit, and restricted publication authority.

11. Assessment and Decision Authority

Editorial assessments and editorial decisions are distinct.

An assessment interprets quality or state.

A decision changes or confirms governed standing.

Decision authority should be proportionate to consequence.

11.1 Ordinary Editorial Decisions

Examples include routine wording revision, duplicate-candidate rejection, low-risk Learning Goal approval, or straightforward alias resolution.

11.2 Elevated Editorial Decisions

Examples include Clinical Case approval, Teaching Relationship activation, Diagnosis Education approval, or governed-artifact supersession.

11.3 High-Risk Editorial Decisions

Examples include emergency management, drug doses, serious-diagnosis exclusion claims, pregnancy or paediatric treatment, and cross-specialty guidance.

11.4 Institutional Decisions

Examples include publication, withdrawal, governance exceptions, diagnosis retirement, and adoption of institutional standards.

12. Editorial Disagreement

Disagreement is an expected feature of responsible clinical governance.

WEOS should not conceal disagreement through silent editing or informal compromise.

A disagreement record should identify:

- disputed issue;
- affected artifact;
- competing interpretations;
- evidence considered;
- clinical and educational consequences;
- reviewers involved;
- unresolved questions;
- required escalation.

The objective is not forced consensus.

The objective is a defensible institutional decision.

13. Escalation and Adjudication

Escalation occurs when ordinary review cannot safely or appropriately resolve an issue.

It may be required because of reviewer disagreement, insufficient expertise, high risk, contradictory evidence, cross-specialty impact, unresolved conflict, or publication urgency.

Adjudication should preserve:

- competing positions;
- evidence considered;
- adjudicating authority;
- final rationale;
- conditions;
- future review obligations.

Adjudication should not erase earlier disagreement.

14. Conflict of Interest

A conflict of interest exists when judgement may be influenced, or reasonably perceived to be influenced, by interests outside the editorial merits of the artifact.

Possible responses include:

- disclosure;
- restricted participation;
- independent review;
- removal from final authority;
- reassignment;
- additional approval.

Undeclared conflicts weaken institutional trust even where the decision is clinically correct.

15. Evidence Oversight

Clinical evidence supports editorial decisions but does not replace them.

Evidence Oversight should ensure that:

- claims are linked to appropriate support;
- sources apply to the stated population and context;
- evidence remains sufficiently current;
- conflicting evidence is represented honestly;
- high-risk claims receive proportionate scrutiny;
- retracted or superseded evidence is identified;
- evidence changes trigger maintenance.

A reference may be credible but not applicable.

Evidence contributes to institutional confidence without independently determining publication.

16. Governance of High-Risk Knowledge

High-risk knowledge may include:

- drug doses;
- emergency treatment;
- procedural guidance;
- contraindications;
- antimicrobial recommendations;
- paediatric and pregnancy management;
- critical-care thresholds;
- diagnostic exclusion statements;
- escalation criteria;
- scoring-system cut-offs;
- guideline-dependent management.

High-risk governance may require stronger evidence, specialty review, independent review, dual approval, explicit scope, stronger validation, shorter maintenance intervals, and defined emergency correction pathways.

Risk should be determined by the consequence of error rather than artifact type alone.

17. Publication Authority

Publication represents an institutional decision to permit governed knowledge to influence learners or institutional activity.

Publication authority must remain separate from authorship, generation, validation, technical deployment, scheduling, and operational availability.

A Publication Authority may publish, schedule, defer, restrict, withdraw, supersede, or decline publication.

The absence of an automated blocker does not constitute publication approval.

18. Withdrawal, Supersession, and Retirement

Published knowledge may require withdrawal, supersession, or retirement.

Withdrawal removes knowledge from active exposure because continued use is inappropriate.

Supersession replaces it with a preferred successor while preserving history.

Retirement ends active editorial use when the artifact or diagnosis no longer serves an appropriate purpose.

Removal from active use must not erase institutional history.

19. Emergency Editorial Governance

Emergency action may be required where learner-facing knowledge presents a credible and significant risk.

Possible actions include immediate withdrawal, temporary suppression, scheduling suspension, disabling playability, restricting generation, or urgent correction.

Every emergency action should preserve authority, risk, time, action, expected duration, and required follow-up.

Urgency may shorten the pathway.

It must not remove accountability.

20. Governance Committees and Editorial Boards

As WEOS grows, governance may be distributed across specialty boards, clinical-safety panels, educational-quality committees, knowledge-governance boards, publication committees, curriculum committees, and evidence groups.

Every governance body should possess:

- defined scope;

- membership requirements;
- decision authority;
- conflict rules;
- preserved decisions;
- escalation pathways;
- review responsibilities.

Committees should not become symbolic approval layers.

21. Specialty Governance

Specialty Governance may define specialty ownership, terminology, reviewer requirements, high-risk categories, evidence expectations, maintenance priorities, and curriculum expectations.

Specialty decisions should remain visible to the wider ecosystem.

Cross-specialty knowledge should support joint review rather than arbitrary ownership.

22. Curriculum Governance

Curriculum Governance evaluates whether the knowledge ecosystem provides coherent learning.

It should consider diagnosis representation, Learning Goal coverage, important mimics, case diversity, learner progression, population representation, specialty balance, and maintenance.

Curriculum governance should not be reduced to counting cases.

Curriculum confidence emerges from educational coherence rather than content volume.

23. Relationship to Editorial Intelligence

Editorial Intelligence may identify blockers, recommend reviewers, detect conflicts, surface evidence deficiencies, prioritise maintenance, calculate readiness, and recommend escalation.

Editorial Intelligence improves governance awareness.

It does not possess governance authority.

A recommendation to approve is not approval.

A readiness result is not publication.

24. Relationship to Artificial Intelligence

Artificial Intelligence may generate, summarise, compare, detect, propose, and prepare context.

It must not independently establish clinical authority, accept evidence, adjudicate disagreement, approve high-risk knowledge, publish, withdraw, or grant exceptions.

AI may assist the decision process.

Human editorial governance remains responsible for the decision.

25. Governance Records

Every material governance action should produce a durable record preserving artifact identity, version, question, actor, authority, evidence, findings, rationale, decision, state change, unresolved concerns, and required follow-up.

A status without rationale is insufficient governance.

A decision without an identifiable decision-maker is insufficient governance.

A changed artifact without version history is insufficient governance.

26. Governance Exceptions

Exceptions should remain rare, explicit, limited, and reviewable.

An exception should identify the requirement, reason, authority, scope, duration, risk, compensating controls, follow-up, and expiry.

Temporary necessity should not become permanent policy through repeated informal use.

27. Governance Quality Assurance

Governance quality may be assessed through completeness of review records, consistency of decisions, evidence sufficiency, disagreement, post-publication correction, unresolved decision age, unauthorised action, emergency-action frequency, overdue maintenance, and separation-of-duty compliance.

WEOS should improve the system producing decisions rather than only correcting individual defects.

28. External Review and Accreditation

External review should be able to determine which standards govern knowledge, who possesses authority, how decisions are recorded, how evidence is assessed, how publication occurs, how errors are corrected, and how AI contributions are governed.

External endorsement may strengthen confidence.

It does not replace internal accountability.

29. Governance Across Institutional Scale

The same principles should remain valid as WEOS grows.



Distributed authority should remain transparent and accountable.

Knowledge approved by one institution should not automatically become approved by every participating institution.

30. Governance Continuity

Governance must remain resilient when people, organisations, and technologies change.

Continuity requires stable identities, transferable ownership, documented authority, preserved decisions, succession, access continuity, restoration capability, maintained standards, and periodic revalidation.

Institutional knowledge should not depend upon the memory of one editor.

31. Principles of Institutional Editorial Governance

Authority Must Be Explicit. Authority should not be inferred solely from authorship, seniority, reputation, or technical access.

Responsibility Must Be Identifiable. Every material decision should have an accountable human decision-maker.

Expertise Must Be Appropriately Scoped. Reviewers should act within their competence and mandate.

Approval Must Apply to a Version. Approval must not apply ambiguously to mutable knowledge.

Publication Requires Separate Authority. Approved knowledge must not become learner-facing without an explicit Publication Decision.

High-Risk Knowledge Requires Stronger Governance. Governance intensity should reflect the consequence of error.

Disagreement Must Remain Visible. Material disagreement should be preserved through resolution.

Conflicts Must Be Managed Transparently. Undisclosed conflicts weaken confidence.

Evidence Supports but Does Not Govern. Sources inform decisions but do not independently create trust.

Technical Capability Does Not Confer Clinical Authority. Administrative and engineering access remain distinct from editorial authority.

Governance Must Remain Auditable. Decisions require context, rationale, responsibility, and history.

Institutional Memory Must Outlive Individuals. Governance should remain durable through personnel and technology change.

32. Relationship to the Wider WEOS Architecture

The Editorial Domain Model defines what is governed.

Editorial Capabilities define what work must be possible.

Editorial Lifecycles define how confidence changes.

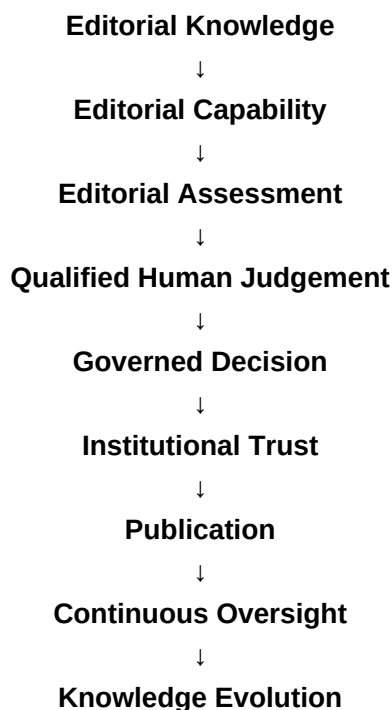
Knowledge Production defines how candidate knowledge enters.

Editorial Intelligence interprets what requires attention.

The Editorial Workspace supports completion of work.

Institutional Editorial Governance defines who may exercise authority, how decisions become trusted, and how accountability is preserved.

The Evolution of the Editorial Knowledge Ecosystem describes how these structures grow.



Closing Principle

Clinical knowledge does not become trustworthy because it was created by an expert, generated by an advanced system, supported by a reference, or made available through software.

It becomes trustworthy when appropriate expertise is exercised through transparent, accountable, evidence-informed, and continuously maintained institutional governance.

People contribute knowledge.

Reviewers evaluate knowledge.

Editorial Intelligence supports judgement.

Governance establishes authority.

Institutions assume responsibility.

Publication expresses trust.

Accountability preserves that trust over time.

Part IX — Evolution of the Editorial Knowledge Ecosystem

1. Purpose

The Wardle Editorial Operating System (WEOS) is designed as an evolving knowledge governance platform.

It is not intended to reach a final state.

Medicine evolves.

Educational practice evolves.

Clinical evidence evolves.

Artificial intelligence evolves.

Editorial governance must therefore evolve alongside them.

The purpose of this chapter is to define how WEOS grows while preserving the architectural principles established throughout this document.

Future evolution should increase the sophistication of editorial governance without changing its philosophical foundations.

2. Philosophy of Evolution

Growth within WEOS should never be measured by the quantity of educational content.

Instead, progress should be measured by the maturity with which clinical knowledge is governed.

As the system evolves it should become:

- more clinically reliable
- more educationally intentional
- more internally consistent
- more explainable
- more scalable
- more maintainable

Evolution therefore represents increasing editorial maturity rather than increasing software complexity.

3. Evolution Through Capability

The evolution of WEOS occurs through progressively richer editorial capabilities.

Each new capability should strengthen one or more existing principles rather than introduce isolated functionality.

Capabilities should evolve in four stages.

Foundational

The editorial concept exists.

Operational

Editors can perform the work.

Governed

Editorial policies consistently govern the work.

Intelligent

Editorial Intelligence proactively assists editors while preserving human authority.

The objective is not automation.

The objective is progressively better editorial judgement.

4. Evolution of Knowledge

Knowledge itself evolves continuously.

Examples include:

- revised evidence
- new diagnostic criteria
- changing terminology
- emerging diseases
- evolving investigations

- changing management recommendations
- improved educational strategies

WEOS should support continuous refinement without disrupting institutional confidence.

Historical governance should always remain preserved.

5. Evolution of Editorial Artifacts

Every governed artifact should be capable of continuous refinement.

A diagnosis may acquire:

- additional cases
- improved objectives
- stronger teaching rules
- refined graph relationships
- improved educational content
- updated evidence

The system should encourage incremental improvement rather than periodic reconstruction.

6. Evolution of Editorial Intelligence

Editorial Intelligence is expected to become increasingly sophisticated throughout the lifetime of WEOS.

Future improvements may include:

- improved dependency reasoning
- richer recommendation models
- stronger quality assessment
- better curriculum analysis
- more comprehensive contradiction detection
- increasingly explainable recommendations
- improved maintenance prioritisation

These improvements strengthen editorial judgement without altering the governance philosophy.

7. Evolution of Artificial Intelligence

Artificial Intelligence will continue to improve.

WEOS should benefit from these advances without becoming dependent upon any specific model, provider, or technology.

Future AI systems may become capable of:

- producing higher quality drafts
- generating richer educational explanations
- identifying increasingly subtle inconsistencies
- proposing stronger knowledge relationships
- analysing curriculum coverage
- assisting reviewers during governance

Regardless of technological advancement, AI remains an editorial contributor rather than an editorial authority.

Human governance remains unchanged.

8. Evolution of Governance

Institutional governance should become progressively more comprehensive.

Examples include:

- richer publication standards
- stronger evidence governance
- improved quality assurance
- expanded curriculum governance
- enhanced specialty governance
- multidisciplinary editorial review
- institutional accreditation support

Governance should expand without compromising transparency or explainability.

9. Evolution of the Editorial Workspace

The Editorial Workspace should evolve according to editorial needs rather than technological trends.

Future workspace improvements should reduce editorial effort while increasing editorial understanding.

As Editorial Intelligence becomes more sophisticated, the workspace should progressively shift from displaying information to supporting judgement.

Editors should spend less time locating knowledge and more time governing it.

10. Institutional Evolution

WEOS is intended to support increasingly sophisticated editorial organisations.

Examples include:

Single editor



Editorial team



Specialty editorial boards



Institutional curriculum committees



Multi-institution collaboration



International editorial networks

The underlying editorial philosophy remains unchanged regardless of organisational scale.

11. Knowledge Network Evolution

As diagnoses mature, the knowledge ecosystem becomes increasingly interconnected.

Future evolution should strengthen:

- differential reasoning
- knowledge relationships
- curriculum integration
- educational continuity
- longitudinal learning

- reasoning pathways

The objective is to develop a coherent network of governed diagnostic knowledge rather than a collection of independent educational resources.

12. Sustainability

WEOS should remain sustainable over decades of editorial development.

Sustainability requires:

- modular editorial artifacts
- independent lifecycles
- transparent governance
- explainable decisions
- maintainable knowledge structures
- preserved editorial history

Architectural decisions should favour long-term maintainability over short-term convenience.

13. Architectural Stability

Although implementation technologies will inevitably change, the conceptual architecture of WEOS should remain stable.

Future implementations may replace:

- user interfaces
- databases
- programming languages
- artificial intelligence models
- storage technologies
- deployment platforms

None of these changes should alter:

- editorial philosophy
- domain model
- governance principles
- lifecycle model
- editorial capabilities
- institutional trust model

These concepts define WEOS independently of implementation.

14. Principles of Evolution

The continued evolution of WEOS should always preserve the following principles.

Knowledge evolves continuously.

Governance evolves continuously.

Editorial Intelligence evolves continuously.

Artificial Intelligence evolves continuously.

Institutional trust evolves continuously.

The architecture should evolve through refinement rather than replacement.

Complexity should increase only when it produces measurable editorial value.

Editorial judgement should always remain central.

15. Long-Term Vision

The long-term vision of WEOS is not simply to become a better editorial platform.

Its vision is to become a comprehensive operating system for governing diagnostic knowledge throughout its entire educational lifecycle.

The mature WEOS should enable institutions to:

- create clinical knowledge
- improve educational quality
- supervise AI-assisted authoring
- maintain evidence-informed content
- preserve institutional memory
- support curriculum development
- continuously strengthen diagnostic education

The value of WEOS ultimately lies not in the software itself, but in the quality of the knowledge it enables institutions to govern.

Closing Principle

The Wardle Editorial Operating System is intentionally designed as an evolving knowledge ecosystem.

Its architecture is founded upon enduring editorial principles rather than transient technologies.

As medicine advances, educational practice develops, and artificial intelligence improves, WEOS should continue to evolve without changing its central purpose:

To enable the transparent, explainable, continuously improving governance of trustworthy clinical knowledge.